PROJECT PROPOSAL

A 3D first-person survival horror series for the web

Boon Sop: The Unbroken Debt

បុណ្យសព៖ បំណុលមិនដាច់

Seven nights, five episodes, one family debt. Khmer folk horror set in French-colonial Cambodia in 1932, playable in any modern desktop browser with no installation.

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01 Executive Summary

Boon Sop: The Unbroken Debt (**បុណ្យសពៈ បំណុលមិនដាច់**) is a first-person atmospheric survival horror game that runs directly in desktop web browsers. It is set in Cambodia in 1932, during the French Protectorate, and draws on Khmer folklore, Buddhist ritual and colonial history.

The player is **Sora**, a 24-year-old illustrator from Phnom Penh. She comes home to Battambang for the funeral of her Uncle Sok and learns that in 1910 her great-grandfather made a pact with a **Neak Ta** (**អ្នកតា**), a guardian spirit of the land: prosperity for the family in exchange for one life from each of four generations. Three lives have been paid. The fourth falls due on the **seventh night** after Sok's death, and Sora discovers that the name written for that final payment is her own.

Across five episodes and seven nights she must break the five seals of the pact, uncover what her uncle really did, and decide what to do about the cousin who would rather she paid.

Project at a glance

GENRE
First-person survival horror
Atmospheric, narrative, stealth and puzzles

PLATFORM
Desktop web browsers
Chrome, Edge, Firefox, Safari

TECHNOLOGY
Babylon.js
WebGPU with WebGL2 fallback

FORMAT
5 episodes
1.5–2.5 hours each · 8–10 hours total

SETTING
Cambodia, 1932
Monsoon season, during Pchum Ben

ENDINGS
Three
Decided by choices across all episodes

Why this game, why now

An unused setting	Horror games regularly draw on Western, Japanese and Korean folklore. Khmer folklore and 1930s Cambodia are almost absent from games.
No-friction access	Browser technology now supports high-quality 3D on ordinary laptops. A link is easier to share than a store page and a large download.
Tight, episodic and achievable	Short, dense episodes respect players' time and keep the project achievable for a solo developer. Episode 1 tests the idea with real players before the rest is built.

02 Vision and Design Pillars

The player should feel like a stranger in her own family's history: alone in the dark, carrying a small light, running out of nights, and surviving through knowledge of rituals and taboos rather than through force.

2.1 Design pillars

Pillar	What it means	How it shows up in play
Cultural authenticity	Khmer architecture, ritual, music and belief are the foundation of the horror, not decoration.	Researched and advisor-reviewed settings; rituals and offerings solve problems.
Light is life	Sora's kerosene lamp is her only defence and her biggest risk.	Oil runs out, light draws some spirits, and shutters and matches force constant choices.
Fully diegetic interface	No on-screen HUD.	Map, notes, inventory and the night count live in Sora's satchel and sketchbook.
Knowledge over force	Sora never fights.	Stealth, sound, line of sight, taboos and counter-rituals replace combat.
Every night is different	No two episodes play the same way.	A new place, spirit and mechanic each episode, plus a structural surprise: a daylight market, a playable flashback, a companion.
Respect is rewarded	The spirit judges how Sora treats the dead, the living and the truth.	Offerings, taboos, compassion and honesty feed directly into the ending.

2.2 Pacing

Each episode is short and dense: roughly **5–6 puzzles, 3–4 escalating threat encounters, one optional side area and one key choice** in about two hours. Every act introduces something new, and every episode ends with a revelation that changes what the player thinks is happening.

03 Story

3.1 The debt

In 1910, famine and cholera swept a forest village in Kampong Cham. Sora's great-grandfather **Chea** went to the village's Neak Ta shrine and offered the family's silver betel box, pleading for his children's lives and for land and fortune. The guardian accepted. The price was **one life from each of four generations**, the fourth falling due one full twelve-year cycle after the third. The pact was bound by five consecrated objects, the **seals**.

The family survived and prospered, moving to rich rice land in Battambang. The guardian collected on schedule. In the 1920s a French rubber company cleared the forest and built its factory over the forgotten shrine.

3.2 The family

Generation	Member	Fate
1	**Chea** ជា · great-grandfather	Made the pact in 1910. First payment: taken days later.
2	**Chhun** ឈុន · grandfather	Second payment: taken in 1911. His widow **Yay Mey** (យាយមី, 66, not of the bloodline) still runs the house.
3	**Kanha** កញ្ញា · Sora's mother	Third payment: drowned in 1920 during a difficult childbirth at a floating clinic on the Tonle Sap. Sora's father Rith took Sora, then 12, to Phnom Penh; he died in 1929.
3	**Sok** សុខ · Kanha's brother	The clinic's medic, who failed to save his sister. Spent twelve years studying the pact. Dies on Night 1.
4	**Vuthy** វុទ្ធី · Sok's son, 26	A clerk for a French trading house who manages the family's land and money. Originally the fourth payment.
4	**Sora** សូរ៉ា · 24	The player. Stopped answering her grandmother's letters after 1929 and never came home. The name now written for the fourth payment.

3.3 Series synopsis

In 1932 the fourth payment falls due. Sok, desperate to save his son, secretly performs a ritual in 1931 that rewrites the debtor's name from **Vuthy** to **Sora**. Riddled with guilt, he then tries to break the seals himself, but only the named debtor can break them, and on Night 1 the guardian strikes him dead. His journal, addressed to "whoever keeps my vigil", is all he leaves behind.

Sora follows the journal, believing she is ending a curse on the family. In **Episode 3** she lives the truth through her uncle's memory: he chose her to die in his son's place, and the drowned spirit haunting the clinic is her own mother. Sok gave her the burden, and also the only way out: if the named debtor breaks all five seals and performs the Ritual of Severance, the debt ends and no one else has to die.

Ending the pact has a cost. "The guardian takes back what it gave." The family's land and fortune would return to the earth. And **Vuthy**, told by Yay Mey that his was the first name written, believes that if Sora tries and fails, the guardian will come for him instead. He would rather she simply paid.

The series is about **what we owe the land and each other**. Chea, the colonial company, Sok and Vuthy all take without paying; Sora, who ran from her family, must decide whether to give back. The full theme, rules of the debt and character arcs are in the *Story Bible*.

3.4 The seven nights

The game takes place over seven nights in the monsoon season of 1932, during the fortnight of Pchum Ben. The guardian collects on the seventh night after Sok's death.

Night	Episode	What happens
1	Prologue (Episode 1)	Sok dies at 8:12 PM trying to burn the first seal. Yay Mey telephones Sora.
3	1 · The House of Mourning	Sora keeps the funeral vigil and breaks Seal #1.
4	2 · The Whispering Pagoda	The last night of Pchum Ben. Seal #2.
5	3 · The Drowned Village	The truth about Sok and Kanha. Seal #3.
6	4 · The Rubber Plantation	The shrine under the factory; Vuthy's betrayal. Seal #4.
7	5 · The Sunken Sanctuary	The Ritual of Severance before dawn.

3.5 The antagonist: the Neak Ta

The Neak Ta appears as a towering shadow wearing an ancient brass mask, growing stronger as each seal breaks. In Khmer belief, Neak Ta are respected guardians of the land, and this one is not evil: it is enforcing a contract the family entered willingly and has tried to cheat twice. It watches how Sora treats the dead, the living and the truth. The true ending reflects this: the guardian bows and releases the family rather than being destroyed.

04 Episode Roadmap

Episode	Setting	Core mechanic and twist on the formula	Spirit	Seal
1 The House of Mourning ផ្ទះបុណ្យ	Stilt house, Battambang · Night 3	Resource management (kerosene, incense, matches). Ends with Sora finding her own name in Sok's chest.	Neak Ta shadow	Bronze Token
2 The Whispering Pagoda វត្តខ្សឹបខ្សៀវ	Village pagoda · Night 4	Sound-based stealth. *Formula break:* a playable 1910 flashback as Chea.	Pret (hungry ghosts)	Silver Betel Box
3 The Drowned Village ភូមិលិចទឹក	Floating village and clinic, Tonle Sap · Night 5	Water and warmth. *Formula break:* opens in daylight at a floating market. *Mid-season twist.*	Chhlong Tonle (Sora's mother)	Terracotta Water Jar
4 The Rubber Plantation ចំការកៅស៊ូ	Colonial estate, Kampong Cham · Night 6	Fog navigation and the "don't look back" taboo. *Formula break:* Vuthy travels with Sora, then betrays her.	Khmaoch Prey (voice mimics)	Iron Tapping Knife
5 The Sunken Sanctuary ប្រាសាទលិចទឹក	Phnom Kulen catacombs · Night 7	Four elemental trials, each with an ancestor's memory, then the Ritual of Severance and the final choice.	Neak Ta Guardian	Golden Inscription Slab

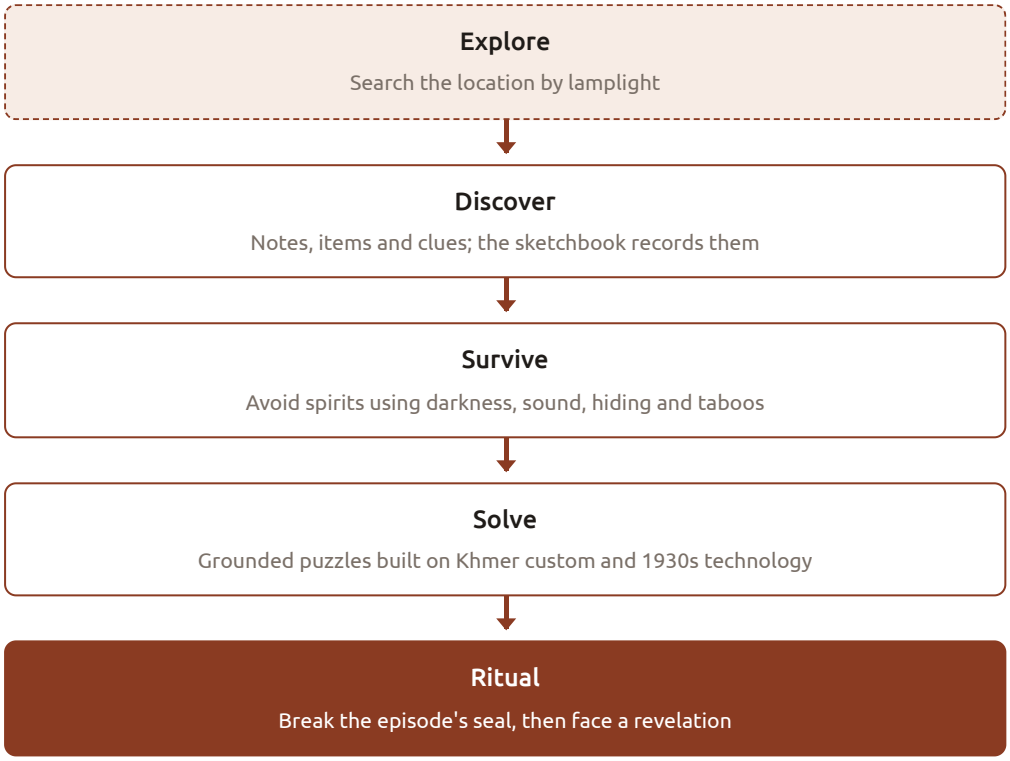
4.1 Episode summaries

Ep 1 · The House of Mourning	Sora keeps the night vigil over Uncle Sok's coffin: lamp fuelled, incense burning, protective thread unbroken. His journal reveals the pact and the seven-night deadline. She burns the Bronze Token before 3:00 AM, then finds a slip of paper in Sok's hand with her own name beside the words "the fourth".
Ep 2 · The Whispering Pagoda	On the last night of Pchum Ben, blind hungry ghosts hunt by sound among the stupas. In the scripture library, Sora relives the 1910 pact as Chea himself. At the 4:00 AM bell she returns the betel box to her ancestors' stupa and the seal breaks.
Ep 3 · The Drowned Village	A daylight floating market gives way to a monsoon night. In the abandoned clinic, Sora learns that Sok renamed the debt to spare his son, and that the water spirit is her mother. Someone has cut her boat loose.
Ep 4 · The Rubber Plantation	By telephone, Yay Mey confesses she knew. Vuthy arrives to "help". In the fog, the mimics speak with his voice too. The factory stands on the 1910 shrine. Vuthy tries to stop the seal from breaking and is seized by the guardian. Sora must choose whether to save him.

Ep 5 · The Sunken Sanctuary	Beneath Phnom Kulen, four trials each end with a memory of one generation of the debt. At the altar, Sora performs the Ritual of Severance while the guardian attacks, and decides Vuthy's fate.

05 Core Gameplay

5.1 Core loop



5.2 Key systems

System	Description
Light and resources	Kerosene lamp, limited matches, and later a shutter that hides the light without putting out the flame.
Diegetic sketchbook	Sora draws maps, copies inscriptions, writes notes and marks the nights remaining.
Stealth and perception	Spirits sense by sight, sound or both. Surfaces change footstep noise; crouching, breath and distractions help.
Escalation	Each episode's spirit grows more dangerous act by act: more of them, sharper senses, new ways to reach Sora.
Cultural taboos	Rules from Khmer belief, such as never answering a voice calling your name in the forest at night. Breaking them has consequences now and at the end.
Puzzles	Physical and cultural: a betel offering tray, temple bells, Khmer lunar dates, a French–Khmer cipher, mirrors, fuses, gears.
Optional areas	One per episode, holding lore notes that teach the chants needed in the finale.
Choices	One key choice per episode, about honesty, compassion and Vuthy.

06 Choices and Endings

Three things are tracked across all five episodes:

Tracked	Built from
Respect	Keeping taboos, making offerings correctly, choosing the honest memory in Episode 2, freeing Sora's mother in Episode 3.
Knowledge	Lore notes found. They teach the correct chant lines used in the final ritual.
Vuthy	Sharing Sok's journal with him (Episode 1), saving him (Episode 4), forgiving him (Episode 5).

Ending	Conditions	Outcome
The Unbroken Freedom true ending	Chants correct, high respect, Vuthy saved and forgiven	Sora and Vuthy offer their blood together. The guardian bows and dissolves; the debt ends with no further death. The family loses its fortune and keeps each other.
The Last Payment bitter ending	Chants correct, but respect too low or Vuthy not saved or not forgiven	The pact is fulfilled rather than severed: the guardian takes Vuthy as the fourth life. Sora lives and the family stays rich, at a price she will carry forever.
Bound to the Soil bad ending	Too many chant errors	The severance fails. Sora wakes inside the coffin at the Battambang stilt house, trapped in the spirit world as her family mourns her.

07 Cultural Foundation

Area	Elements used
Spirits and beliefs	Neak Ta (អ្នកតា), Pret (ព្រេត), Khmaoch Prey (ខ្មោចព្រៃ), a drowned water spirit
Rituals and time	Funeral vigil and the seventh-day rites, Pchum Ben (ភ្ជុំបិណ្ឌ), Bay Ben (បាយបិណ្ឌ), betel offerings, protective cotton thread, incense, chanting, the Khmer lunar calendar and twelve-year animal cycle
Architecture	Teak stilt houses (ផ្ទះពោងជើង), pagodas, stupas (ចេតិយ), floating villages, French-colonial estates, Angkorian sandstone
Objects and texts	Palm-leaf manuscripts (ស្លឹករឹត), betel boxes, terracotta jars, brass ritual objects, Khmer numerals
Music	Pinpeat (ពិណពាទ្យ), especially the roneat (រនាត)
History	The French Protectorate, rubber plantations and their labourers, famine and epidemic, rural and lake life

08 Technology and Performance

Layer	Choice
Engine	Babylon.js (TypeScript): WebGPU with automatic WebGL2 fallback
Physics	Havok (bundled with Babylon.js)
3D and textures	glTF 2.0 (`.glb`) with Draco/meshopt compression; KTX2 / Basis Universal textures
Audio	Web Audio API through Babylon.js Audio Engine v2: 3D spatial sound and reverb
Saves	IndexedDB plus portable save codes that carry choices between episodes
Build and hosting	Vite, deployed as a static website

Performance target	Value
Frame rate	60 fps at 1080p (dedicated GPU) · 30 fps at 720p (integrated laptop GPU)
Draw calls / visible polygons	120 or fewer / 150,000 or fewer per frame
Initial download per episode	85 MB or less, first playable area under 20 MB

Chrome and Edge use WebGPU; Safari on macOS 26+ and Firefox on Windows and Apple Silicon Macs use WebGPU; everything else falls back to WebGL2. Mobile is not supported in the first release.

09 Production Plan

A **solo developer** builds the game, using licensed libraries for generic assets and custom work for everything culturally specific. **Episode 1 is built and released first.**

Phase	Name	Length	Main outcome
0	Pre-production	3 weeks	Beat sheet, art bible, tools installed, advisors found
1	Technical spikes	4 weeks	Risky technology proven
2	Greybox	4 weeks	All of Episode 1 playable with placeholder shapes
3	Core systems	7 weeks	Reusable, data-driven systems
4	Art and audio	8 weeks	Final environments, characters, sound and lighting
5	Polish and release	5 weeks	Playtesting, cultural review, browser testing, release

That is about **31 weeks to the Episode 1 release**, then about 3–4 months per later episode, for a total of roughly 2–2.5 years at about 25 hours a week. Details are in the *Development Plan*.

10 Risks and Mitigation

Risk	Impact	Mitigation
Scope too large for one person	Critical	Short, dense episodes; release Episode 1 first; build systems once; keep a cut list.
Story complexity across episodes	High	One series outline (this document) that all episode documents follow; choices stored in saves from Episode 1.
Browser memory and loading	High	Per-area loading, KTX2 textures, streamed audio, memory profiling every test cycle.
Laptop performance	High	Weekly tests on integrated graphics; one shadowed light; baked lighting; quality presets.
Cultural misrepresentation	High	Cambodian advisors review the story, the greybox and the release candidate; native-speaker language checks.
Browser differences	Medium	Always keep the WebGL2 fallback; test all four browsers before each release.

11 Conclusion and Next Steps

Boon Sop: The Unbroken Debt brings Khmer folklore to a global horror audience through the most accessible platform there is. Its seven-night countdown, betrayal at the heart of the family and choice-driven endings give a short series real weight.

1. Review this version of the story with one or two Cambodian cultural advisors.
2. Write the Episode 1 beat sheet and a one-page art bible.
3. Install the toolchain and run the technical spikes.
4. Build the Episode 1 greybox and playtest it.

A Appendix: Document Set and Revision History

No.	Document	Purpose
00	Project Proposal	Vision, story, scope and plan overview
01–05	Episode specifications	Full design and narrative specification for each episode
06	Development Plan	Roadmap, architecture, systems, assets, resources and open questions
07	Story Bible	Theme, rules of the debt, character arcs, series structure, timeline
08–12	Phase 0 production documents	Episode 1 beat sheet and level layout, art bible, Episode 1 script, technical design document, controls and interface
13–15	Production documents	Audio design document, cultural guide and glossary, playtest plan; plus the asset tracker spreadsheet

Version	Changes
1.0	Original proposal and episode documents.
2.0	Professional rewrite and reorganisation; Babylon.js chosen; solo production plan; Khmer text fixed.
3.0	Shorter, denser episodes (8–10 hours total). New story layer: the four-payment debt, Sora's family, Sok's renaming twist, cousin Vuthy, the seven-night countdown, the 1910 shrine under the estate, three endings, escalation, new cultural puzzles, optional areas and one key choice per episode.
3.1	Story Bible added. Theme set; rules of the debt fixed; Sora given a flaw and an artistic climax; Yay Mey's confession; Vuthy's motive changed from money to fear; major reveals made playable or audible instead of read.